

Zero-Shot Topic Labeling: Extensions and Model Comparison

Supervisor Meeting - Experiment-1

Daniel Tesfai

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1 Starting Point

- Base paper (BTW2025-122): zero-shot topic labeling with Llama 3 8B, 19-topic vocabulary, 2500 papers.
- Reported baseline: 50% success, hallucination 25% (titles) / 33% (abstracts).
- Two extensions investigated: (1) fixing the matching step, (2) testing newer models and model size.

2 Extension 1: Matching Fix

- Original matching: exact string match only, breaks on quotes or near-miss wording.
- Fix: strip punctuation, then word-containment fuzzy match, with the ontology's parent category (`computer science`) explicitly excluded from auto-resolution.

Input	Old matching	New matching
"'software engineering'"	hallucination	software engineering
computer vision	hallucination	computer imaging and vision
computer science	hallucination	hallucination (correct)

At full scale (Qwen3-8B, titles): 103/2500 answers (4.1%) recovered by the fix, all the literal string `computer vision`, 83 became successes, 20 became misclassified, 0 stayed unresolved.

3 Extension 2: Newer Models vs. Paper Baseline

Model	Success	Misclassified	Hallucination
Paper (Llama 3 8B), titles	50%	25%	25%
Paper (Llama 3 8B), abstracts	50%	17%	33%
Qwen3-8B, titles	68.2%	29.3%	2.5%
Qwen3-8B, abstracts	70.0%	27.2%	2.8%
Claude Haiku 4.5, titles	70.9%	27.6%	1.5%
Claude Haiku 4.5, abstracts	74.0%	23.3%	2.7%

- Every newer model beats the paper by a wide margin, driven mainly by hallucination collapsing to under 4%.
- Titles vs. abstracts: at native precision, Qwen3-8B and Claude both favor abstracts, matching every Qwen3.5 size tested in Section 4. An earlier GGUF-quantized run of Qwen3-8B had shown the opposite (titles ahead), which in hindsight looks like a precision artifact of that specific backend, not a genuine per-model difference.

4 Extension 3: Does Model Size Matter?

Same task, same prompt, Qwen3.5 family at five sizes (0.8B to 27B).

Size	Titles	Abstracts
0.8B	52.1%	53.7%
4B	75.4%	76.7%
9B	78.6%	80.8%
27B	76.0%	77.0%

- **9B achieves the highest accuracy**, not the largest model tested. Scaling helps sharply to 4B, a little more to 9B, then reverses at 27B.
- 0.8B already beats the paper’s Llama 3 8B baseline (52.1% vs 50%), a model ten times its size, from an older generation.

5 Bonus Finding: Quantization Sensitivity Scales With Size

- **What:** storing and computing a model's weights with fewer bits than the precision it was trained in. Here: 16-bit floating point (bf16) reduced to 4-bit. A lossy compression of the model's parameters, not the model itself.
- **Why:** roughly 4x less memory per parameter. This is what let the 27B checkpoint (needs ~54GB in bf16) fit entirely on a single 24GB GPU, avoiding a much slower fallback where the excess spills into system RAM. It can also increase inference speed, since less data has to move through memory per token generated.

Size	Field	bf16 (default)	4-bit	Difference
0.8B	titles	52.1%	39.2%	-12.9 pts
0.8B	abstracts	53.7%	27.6%	-26.1 pts
4B	titles	75.4%	74.6%	-0.8 pts
9B	titles	78.6%	80.0%	+1.4 pts
27B	titles	75.4%	76.0%	+0.6 pts

- Small models are fragile under 4-bit quantization, 0.8B loses over a quarter of its accuracy edge.
- 4B and up are effectively indifferent, gains and losses under 1.5 points.

6 Full Results Table

Run	Success	Misclassified	Hallucination
Paper (Llama 3 8B), titles	50%	25%	25%
Paper (Llama 3 8B), abstracts	50%	17%	33%
Qwen3.5-0.8B, titles	52.1%	47.3%	0.6%
Qwen3.5-0.8B, abstracts	53.7%	43.9%	2.4%
Qwen3-8B, titles	68.2%	29.3%	2.5%
Qwen3-8B, abstracts	70.0%	27.2%	2.8%
Claude Haiku 4.5, titles	70.9%	27.6%	1.5%
Claude Haiku 4.5, abstracts	74.0%	23.3%	2.7%
Qwen3.5-4B, titles	75.4%	24.0%	0.6%
Qwen3.5-4B, abstracts	76.7%	22.3%	1.0%
Qwen3.5-9B, titles	78.6%	20.8%	0.6%
Qwen3.5-9B, abstracts	80.8%	18.7%	0.5%
Qwen3.5-27B, titles (4-bit)	76.0%	24.0%	0.04%
Qwen3.5-27B, abstracts (4-bit)	77.0%	22.9%	0.1%

7 Titles vs. Abstracts

- Abstracts beat titles in every single run tested, across all five Qwen3.5 sizes, Qwen3-8B, and Claude Haiku 4.5, no exceptions.
- The gap is small and fairly stable, roughly 1–3 points, smallest for the 27B model (+1.0 pt) and largest for Claude Haiku 4.5 (+3.1 pts). More input text helps a little across the board, but it never changes which model or size wins.

8 Conclusion

- **Extension 1 (matching fix):** mainly moves the hallucination column, not the model’s actual answers. 4.1% of Qwen3-8B title answers were wrongly counted as hallucination due to fragile exact-string matching. The fix recovered most of them into success.
- **Extension 2 (newer models):** mainly collapses hallucination, from 25–33% (paper) to under 4% for every newer model tested, while success rises 17–30 points and misclassified stays roughly flat or rises slightly.

- **Extension 3 (model size):** hallucination stays low and flat across every size tested (0.6% to 2.4%, no trend with size), so hallucination is a generation/training effect, not a size effect. What moves with size is the success/misclassified split: 0.8B misclassifies nearly half the time (47.3%/43.9%), 9B misclassifies about a fifth of the time. Size affects whether the model picks the *correct* topic, not whether it stays inside the vocabulary, and this effect does not improve indefinitely, it peaks at 9B and reverses at 27B.
- **Taken together:** the three extensions target different parts of the original failure mode. The matching fix and newer models both attack hallucination, one through evaluation robustness, the other through instruction-following, while model size mainly shifts the success/misclassified balance, that is, topic judgment, with diminishing and eventually reversing returns.

9 Open Questions for Discussion

- Which direction next: cost optimization and multi-label evaluation, or few-shot prompting?